

SKILL 17

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <signal.h>
#include <string.h>

typedef struct {
    int job_id;
    pid_t pid;
    char command[50];
    char status[20];
} Job;

Job job;

void show_job() {
    printf("\n=== Active Job List ===\n");
    printf("[%d] PID: %d | Command: %s | Status: %s\n",
        job.job_id,
        job.pid,
        job.command,
        job.status);
}
```

```
int main() {
    printf("=== Task 17: Job Listing and Foreground Switching ===\n");

    job.job_id = 1;
    strcpy(job.command, "sleep 10");

    pid_t pid = fork();

    if (pid < 0) {
        perror("fork");
        return 1;
    }

    if (pid == 0) {
        sleep(10);
        exit(0);
    }

    job.pid = pid;
    strcpy(job.status, "Running");

    printf("\nBackground job created successfully\n");

    show_job();

    printf("\n=== Retrieve Job ID ===\n");
    printf("Selected Job ID: %d\n", job.job_id);
    printf("Selected PID: %d\n", job.pid);

    printf("\n=== Stop Job Test ===\n");

    kill(job.pid, SIGSTOP);

    strcpy(job.status, "Stopped");

    show_job();

    printf("\n=== Foreground Switching Test ===\n");
    printf("Target job identified: Job %d\n", job.job_id);

    printf("Resuming process...\n");

    kill(job.pid, SIGCONT);

    strcpy(job.status, "Running");
}
```

```

printf("\n=== Foreground Switching Test ===\n");
printf("Target job identified: Job %d\n", job.job_id);

printf("Resuming process...\n");

kill(job.pid, SIGCONT);

strcpy(job.status, "Running");

show_job();

printf("\nWaiting for foreground job to complete...\n");

waitpid(job.pid, NULL, 0);

strcpy(job.status, "Completed");

printf("\n=== Updated Job State ===\n");

show_job();

printf("\n=== Job Listing Test Complete ===\n");
printf("Job ID retrieved successfully\n");
printf("Status information updated successfully\n");
printf("Foreground process completed successfully\n");

printf("\n=== Task 17 Complete ===\n");

return 0;
}

```

Output:

=== Task 17: Job Listing and Foreground Switching ===

Background job created successfully

=== Active Job List ===

[1] PID: 520 | Command: sleep 10 | Status: Running

=== Retrieve Job ID ===

Selected Job ID: 1

Selected PID: 520

=== Stop Job Test ===

=== Active Job List ===

[1] PID: 520 | Command: sleep 10 | Status: Stopped

=== Foreground Switching Test ===

Target job identified: Job 1

Resuming process...

=== Active Job List ===

[1] PID: 520 | Command: sleep 10 | Status: Running

Waiting for foreground job to complete...

=== Updated Job State ===

=== Active Job List ===

[1] PID: 520 | Command: sleep 10 | Status: Completed

=== Job Listing Test Complete ===

Job ID retrieved successfully

Status information updated successfully

Foreground process completed successfully